

CURRICULUM VITAE
University of Pittsburgh
School of Medicine

BIOGRAPHICAL

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EDUCATION AND TRAINING

UNDERGRADUATE

Dates Attended	Institution	Degree Received	Major Subject
2003–2010	Soongsil University Seoul, South Korea	B.S., 2011	Chemical Engineering

GRADUATE

Dates Attended	Institution	Degree Received	Major Subject
2013–2014	Carnegie Mellon University, Pittsburgh, PA	M.S., 2014	Chemical Engineering
2016–2018	University of Ulsan College of Medicine, Seoul, South Korea	Ph.D. coursework completed Advisor: Buhm Han, Ph.D.	Biomedical Science
2019–2021	Seoul National University College of Medicine, Seoul, South Korea	Ph.D., 2021 Advisor: Buhm Han, Ph.D.	Biomedical Science

POSTGRADUATE

Dates Attended	Institution	Program Director	Discipline
2015–2016	Research Associate	Buhm Han, Ph.D.	Bioinformatics

Asan Medical Center, Asan
Institute for Life Science, Seoul,
South Korea

2018–2019	Research Associate, Seoul National University, Seoul, South Korea	Buhm Han, Ph.D.	Statistical Genetics
2021–2026	Postdoctoral Research Scientist, Columbia University, Department of Biostatistics, New York, NY	Iuliana Ionita-Laza, Ph.D.	Statistical Genetics & Genetic Epidemiology

APPOINTMENTS AND POSITIONS

ACADEMIC

Years	Institution	Position
2026– Present	University of Pittsburgh School of Medicine, Department of Psychiatry, Pittsburgh, PA	Assistant Professor, Tenure Stream

MEMBERSHIPS IN PROFESSIONAL AND SCIENTIFIC SOCIETIES

Year	Organization
2015–Present	American Society of Human Genetics
2023–Present	International Genetic Epidemiology Society

HONORS

Year	Title of Award
2016	Graduate Student Scholarship Korea Institute of Science and Technology Information
2016–2018	Graduate Scholarship Recipient University of Ulsan College of Medicine
2017	Presentation Award Korea Genome Organization

2017	The Paper of the Year University of Ulsan College of Medicine
2018	Best Presentation Award Korea Genome Organization
2019	Excellent Poster Award Korea Genome Organization
2019–2021	Lecture & Research Scholarship Seoul National University College of Medicine
2019–2021	Graduate School Scholarship in Biomedical Sciences Asan Foundation (\$20,000/year)
2021	Graduate with Distinction (GPA: 4.5/4.5) Seoul National University College of Medicine
2025	Neal Award Finalist International Genetic Epidemiology Society (IGES)

PUBLICATIONS

ORIGINAL PEER REVIEWED ARTICLES

2015

1. Kwak SH, Kim YJ, Chae J, **Lee CH**, Han B, Kim JI, Jung HS, Cho YM, Park KS. Association of HLA Genotype and Fulminant Type 1 Diabetes in Koreans. *Genomics Inform.* 2015;13(4):126-31. Epub 20151231. doi: 10.5808/GI.2015.13.4.126. PubMed PMID: 26865843; PubMed Central PMCID: PMC4742322.

2016

2. Han B, Pouget JG, Slowikowski K, Stahl E, **Lee CH**, Diogo D, Hu X, Park YR, Kim E, Gregersen PK, Dahlqvist SR, Worthington J, Martin J, Eyre S, Klareskog L, Huizinga T, Chen WM, Onengut-Gumuscu S, Rich SS, Major Depressive Disorder Working Group of the Psychiatric Genomics C, Wray NR, Raychaudhuri S. A method to decipher pleiotropy by detecting underlying heterogeneity driven by hidden subgroups applied to autoimmune and neuropsychiatric diseases. *Nat Genet.* 2016;48(7):803-10. Epub 20160516. doi: 10.1038/ng.3572. PubMed PMID: 27182969; PubMed Central PMCID: PMC4925284.
3. Jung J, **Lee CH**, Seol HS, Choi YS, Kim E, Lee EJ, Rhee JK, Singh SR, Jun ES, Han B, Hong SM, Kim SC, Chang S. Generation and molecular characterization of pancreatic cancer patient-derived xenografts reveals their heterologous nature. *Oncotarget.* 2016;7(38):62533-46. doi:

10.18632/oncotarget.11530. PubMed PMID: WOS:000387164700128.

4. **Lee CH**, Cook S, Lee JS, Han B. Comparison of Two Meta-Analysis Methods: Inverse-Variance-Weighted Average and Weighted Sum of Z-Scores. *Genomics Inform.* 2016;14(4):173-80. Epub 20161230. doi: 10.5808/GI.2016.14.4.173. PubMed PMID: 28154508; PubMed Central PMCID: PMC5287121.
5. Network NSG, International Stroke Genetics C. Loci associated with ischaemic stroke and its subtypes (SiGN): a genome-wide association study. *Lancet Neurol.* 2016;15(2):174-84. Epub 20151219. doi: 10.1016/S1474-4422(15)00338-5. PubMed PMID: 26708676; PubMed Central PMCID: PMC4912948.
6. van 't Hof FN, Ruigrok YM, **Lee CH**, Ripke S, Anderson G, de Andrade M, Baas AF, Blankensteijn JD, Bottinger EP, Bown MJ, Broderick J, Bijlenga P, Carrell DS, Crawford DC, Crosslin DR, Ebeling C, Eriksson JG, Fornage M, Foroud T, von Und Zu Fraunberg M, Friedrich CM, Gaal EI, Gottesman O, Guo DC, Harrison SC, Hernesniemi J, Hofman A, Inoue I, Jaaskelainen JE, Jones GT, Kiemenev LA, Kivisaari R, Ko N, Koskinen S, Kubo M, Kullo IJ, Kuivaniemi H, Kurki MI, Laakso A, Lai D, Leal SM, Lehto H, LeMaire SA, Low SK, Malinowski J, McCarty CA, Milewicz DM, Mosley TH, Nakamura Y, Nakaoka H, et al. Shared Genetic Risk Factors of Intracranial, Abdominal, and Thoracic Aneurysms. *J Am Heart Assoc.* 2016;5(7). Epub 20160714. doi: 10.1161/JAHA.115.002603. PubMed PMID: 27418160; PubMed Central PMCID: PMC5015357.
7. Yang SK, Hong M, Oh H, Low HQ, Jung S, Ahn S, Kim Y, Baek J, **Lee CH**, Kim E, Kim KM, Ye BD, Kim KJ, Park SH, Lee HS, Lee I, Shin HD, Han B, McGovern DPB, Liu JJ, Song K. Identification of Loci at 1q21 and 16q23 That Affect Susceptibility to Inflammatory Bowel Disease in Koreans. *Gastroenterology.* 2016;151(6):1096-+. doi: 10.1053/j.gastro.2016.08.025. PubMed PMID: WOS:000390955500023.

2017

8. Kim EE, Lee S, **Lee CH**, Oh H, Song K, Han B. FOLD: a method to optimize power in meta-analysis of genetic association studies with overlapping subjects. *Bioinformatics.* 2017;33(24):3947-54. doi: 10.1093/bioinformatics/btx463. PubMed PMID: 29036405; PubMed Central PMCID: PMC5860085.
9. **Lee CH**, Eskin E, Han B. Increasing the power of meta-analysis of genome-wide association studies to detect heterogeneous effects. *Bioinformatics.* 2017;33(14):i379-i88. doi: 10.1093/bioinformatics/btx242. PubMed PMID: 28881976; PubMed Central PMCID: PMC5870848.

2018

10. Han B, Akiyama M, Kim KK, Oh H, Choi H, **Lee CH**, Jung S, Lee HS, Kim EE, Cook S, Haritunians T, Yamazaki K, Park SH, Ye BD, McGovern DPB, Esaki M, Kawaguchi T, Khor SS, Taylor KD, Rotter JI, Suzuki Y, Matsui T, Motoya S, Bang SY, Kim TH, Momozawa Y, Kamatani Y, Tokunaga K, Kubo M, Okada Y, Yang SK, Song K. Amino acid position 37 of HLA-DRbeta1

affects susceptibility to Crohn's disease in Asians. *Hum Mol Genet.* 2018;27(22):3901-10. doi: 10.1093/hmg/ddy285. PubMed PMID: 30084967.

11. Kang EY, **Lee CH**, Furlotte NA, Joo JWJ, Kostem E, Zaitlen N, Eskin E, Han B. An Association Mapping Framework To Account for Potential Sex Difference in Genetic Architectures. *Genetics.* 2018;209(3):685-98. doi: 10.1534/genetics.117.300501. PubMed PMID: WOS:000437171700005.

2021

12. **Lee CH**, Shi H, Pasaniuc B, Eskin E, Han B. PLEIO: a method to map and interpret pleiotropic loci with GWAS summary statistics. *American Journal of Human Genetics.* 2021;108(1):36-48. doi: 10.1016/j.ajhg.2020.11.017. PubMed PMID: WOS:000606453800005.

2023

13. Jung S, **Lee CH**, Sul JH, Han B. Building an optimal predictive model for imputing tissue-specific gene expression by combining genotype and whole-blood transcriptome data. *HGG Adv.* 2023;4(4):100223. Epub 20230711. doi: 10.1016/j.xhgg.2023.100223. PubMed PMID: 37576186; PubMed Central PMCID: PMC10413136.

2025

14. **Lee CH**, Khan A, Wang C, Weng CH, Buxbaum JD, Kiryluk K, Ionita-Laza I. Liability threshold model-based disease risk prediction based on electronic health record phenotypes. *Nature Genetics.* 2025. doi: 10.1038/s41588-025-02370-4. PubMed PMID: WOS:001606642600001.
15. Luo Y, Khan A, Liu L, **Lee CH**, Perreault GJ, Pomenti SF, Gourh P, Kiryluk K, Bernstein EJ. Cross-Phenotype Genome-Wide Association Study on the Shared Genetic Susceptibility to Systemic Sclerosis and Primary Biliary Cholangitis. *Arthritis Rheumatol.* 2025;77(6):727-39. Epub 20250227. doi: 10.1002/art.43081. PubMed PMID: 39676709; PubMed Central PMCID: PMC12124972.

2026

16. Wu HQ, **Lee CH**, Abiri N, Ionita-laza I. Domain-Aware Matrix Completion for Phenotype Imputation Using Electronic Health Record Data with Applications in Genomic Research. *Ann Appl Stat.* 2026;20(2):1010-32. doi: 10.1214/26-Aoas2165. PubMed PMID: WOS:001819837900006.

OTHER PEER-REVIEWED PUBLICATIONS

2025

17. **Lee CH**. Pleiotropy in Multitrait Genome-Wide Association Studies: Approaches and Insights From Schizophrenia-Cognition Analyses. *Biological Psychiatry.* 2025;98(9):650-1. doi: 10.1016/j.biopsych.2025.08.004. PubMed PMID: WOS:001590312200002.

ABSTRACTS

ORAL/Podium Presentations

2017

1. Increasing the Power of Meta-Analysis of Genome-Wide Association Studies to Detect Heterogeneous Effects. Presented at the 25th Annual International Conference on Intelligent Systems for Molecular Biology and the 16th European Conference on Computational Biology, Prague, Czech Republic, July 2017.

2025

2. covImpute: Leveraging Genetic Correlations to Impute Missing EHR Phenotypes. International Genetic Epidemiology Society Annual Meeting, Cologne, Germany, September 2025.

Poster presentation

2016

3. Comparison of Two Meta-Analysis Methods: Inverse-Variance-Weighted Average and Weighted Sum of Z-Scores. Korea Society for Biochemistry and Molecular Biology.
4. A Powerful Framework for Combining Heterogeneous Data in Genetic Association Studies. American Society of Human Genetics Annual Meeting.

2017

5. Increasing the Power of Meta-Analysis of Genome-Wide Association Studies to Detect Heterogeneous Effects. Korea Genome Organization Annual Meeting, South Korea.
6. Increasing the Power of Meta-Analysis of Genome-Wide Association Studies to Detect Heterogeneous Effects. American Society of Human Genetics Annual Meeting.
7. Increasing the Power of Meta-Analysis of Genome-Wide Association Studies to Detect Heterogeneous Effects. University of Ulsan College of Medicine, South Korea.

2018

8. A Framework Optimized for Detecting Pleiotropic Loci of Correlated Traits. Korea Genome Organization Annual Meeting, South Korea.
9. An Association Mapping Framework to Account for Potential Sex Differences in Genetic Architectures. American Society of Human Genetics Annual Meeting.

2019

10. An Association Mapping Framework to Account for Potential Sex Differences in Genetic Architectures. Korea Genome Organization Annual Meeting, South Korea.

2020

11. Analysis of 18 Cardiovascular Disease-Related Traits. Korea Genome Organization Annual Meeting, South Korea.

2021

12. PLEIO: A Method to Map and Interpret Pleiotropic Loci Using GWAS Summary Statistics. Korea Genome Organization Annual Meeting, South Korea.

2022

13. LTPI: A Generalized Liability Threshold Model Combining Dichotomous and Continuous Phenotypes in Electronic Health Records Increases Association Power. American Society of Human Genetics Annual Meeting.

2023

14. Liability Threshold Model-Based Disease Risk Prediction Using Electronic Health Record Phenotypes. American Society of Human Genetics Annual Meeting.
15. Liability Threshold Model-Based Disease Risk Prediction Using Electronic Health Record Phenotypes. International Genetic Epidemiology Society Annual Meeting.

2025

16. covImpute: Leveraging Genetic Correlations to Impute Missing Electronic Health Record Phenotypes. American Society of Human Genetics Annual Meeting.

2026

18. GLEN: A Deep Learning Framework for Genetic Liability Prediction from Multimodal Data. American Society of Human Genetics Annual Meeting.
19. GLEN: A Deep Learning Framework for Genetic Liability Prediction from Multimodal Data. International Genetic Epidemiology Society Annual Meeting.

PROFESSIONAL ACTIVITIES

TEACHING

Mentoring

2020–2021	Research Mentor, Yeonsun Yang, Undergraduate Student Mentored undergraduate research during a spring internship at Seoul National University on predicting tissue-specific gene expression using genotype and whole-blood transcriptome data.
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RESEARCH

Current Grant Support

Grant Number	Grant Title	Role	Years	Source/ \$Amount
1K99HG013546 (K99 Phase)	Multi-Trait Analysis in Large-Scale Biobank Datasets Linked to Electronic Health Records	PI	2024–2026	NHGRI/ \$250,000

Pending Grant Support

Grant Number	Grant Title	Role	Year	Source/ \$Amount
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4R00HG013546 (R00 Phase)	Multi-Trait Analysis in Large-Scale Biobank Datasets Linked to Electronic Health Records	PI	2027– 2029	NHGRI/ \$750,000
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INVITED SEMINAR AND LECTURESHIPS

Local Presentations

March 2025 — My Journey to K99: Research Strategy and Innovation Social Lunch. Office of the Vice Dean for Research Strategy, Columbia University, New York, NY.

National Presentations

February 2025 — Harnessing Pleiotropic Signals to Transform Precision Medicine. Cancer Control and Population Science Program Meeting, University of New Mexico Comprehensive Cancer Center, Albuquerque, NM.

December 2025 — Multi-Trait Genetic Inference Using Latent Risk Modeling. Center for Disease Neurogenomics, Icahn School of Medicine at Mount Sinai, New York, NY.

December 2025 — From Phenotypic Correlation to Genetic Liability: Modeling Disease Risk with LTPI. Division of Epidemiology, Vanderbilt University, Nashville, TN.

January 2026 — Rethinking Germline Cancer Risk Modeling Using Population Genomics and Electronic Health Records. Division of Molecular Medicine, University of New Mexico Comprehensive Cancer Center, Albuquerque, NM.

February 2025 — Harnessing Pleiotropic Signals to Transform Precision Medicine. Cancer Control and Population Science Program Meeting, University of New Mexico Comprehensive Cancer Center, Albuquerque, NM.

March 2026 — Integrating Genomics and EHR Data to Improve Disease Risk Prediction. University of Oklahoma Health Campus, Oklahoma City, OK.

International Presentations

January 2026 — Medical Record-Based Disease Risk Prediction Model: LTPI. Academic Seminar, Biological Research Information Center, Pohang University of Science and Technology, Pohang, South Korea.

February 2026 — Integrating EHR Phenotypes and Genomics for Population-Scale Risk Estimation. Graduate School of Medical Science and Engineering, Korea Advanced Institute of Science and Technology, Daejeon, South Korea.

SERVICE

Years	Institution	Context
2026– Present	Director of Statistical Genetics in Psychiatry	Department of Psychiatry, University of Pittsburgh School of Medicine; leads statistical genetics and collaborative genomic research.